Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/02/2006
PI Name:	Li, Ben Q. Ph.D.		
Project Title:	Modeling Study of Instability and Internal Flows in Levitated Droplets Under Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	benqli@engin.umd.umich.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	313 593 5465
Organization Name:	University of Michigan-Dearborn		
PI Address 1:	Dbn-Col Engr & Comp Sci		
PI Address 2:	1344 EC 1491		
PI Web Page:			
City:	Dearborn	**State:**	MI
Zip Code:	48128	**Congressional District:**	15
Comments:	Moved to University of Michigan, per D. Gillies/MSFC (4/07), from Washington State University.		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Banks, Curtis	**Contact Phone:**	256 544-4437
Contact Email:	curtis.e.banks@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNM04AA17G		
Performance Goal No.:			
Performance Goal Text:			

Numerical models will be developed and extensive numerical simulations will be carried out to determine the critical Reynolds number above which the melt flow instability exacerbates itself and the mechanism for flow instability in magnetically levitated droplets. The results are then fed into numerical models for direct numerical simulations of turbulence and the statistic characteristics of turbulence are then determined, which include turbulent energy spectra (both spanwise and streamwise) and energy budget (production, dissipation and turbulent diffusion), mean velocity distribution, Reynolds stress distributions, skewness and flatness of fluctuating quantities, turbulent intensities, vorticity distributions and dynamics, integral length scales and instantaneous flow structures. The computed results, along with available experimental data, will allow us to elucidate the basics governing turbulent magnetically-driven flows in magnetically levitated drops and will be used also to determine the coefficients for the engineering turbulence models, which so far have been based on wall-bound flow studies. These engineering turbulence models are then incorporated

Task Description:	into the numerical models already developed by us for the electrodynamic, fluid flow and heat transfer, free surface deformation and surface oscillation phenomena associated with magnetically levitated droplets. Three-dimensional numerical models will be developed for magnetically and electrostatically levitated droplets and simulations will be carried out to study the axial, off-axis rotational and spin instabilities that may be attributed to the disturbance of the space environment. Three-dimensional model will also be developed for the Marangoni flows in electrostatically levitated droplets that result from different arrangements of heating sources. Experiments of Marangoni flows in electrostatically levitated droplets will be conducted using a laser PIV system and measurements will be used to verify and refine model predictions. The objective of the proposed research is to develop numerical models for direct numerical simulation of turbulence in levitated droplets in magnetic fields, to carry out extensive numerical simulations and statistical analysis of turbulence in the droplets, to characterize the k-e based engineering models using the database obtained from the direction numerical simulations and to identify the turbulence model most suitable for describing turbulent flows in droplets, use these models to study laminar and turbulent transition in levitated droplets. To study the stability of melt flows in the droplets and study the levitation stability of levitated droplets. The proposed research effort addresses critical issues concerning the use of free droplets for the fundamental understanding of phase transformation and kinetics that requires long-duration, high quality microgravity conditions. The proposed work will be built upon our substantial drop modeling effort undertaken thus far from NASA's funding. It is believed that the proposed research will generate information and useful tools critical not only for an in-depth understanding of the fundamentals governing the behavior of magnetically or electrostatically levitated drops, but also for design and improvement of magnetic and/or electrostatic levitation facilities for space applications, for rational planning of microgravity experiments with magnetically or electrostatically levitated droplets and interpreting the measured results obtained from space experiments for phase transformation and kinetics studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A full 3-D model for steady state and transient Marangoni fluid flow in deformed droplet levitated in electrostatic levitator has been developed. Extensive simulations have been carried out using the 3-D model for a variety of levitation conditions under both earthbound and microgravity conditions. The results obtained from the model have been published in archive journals and presented in national/international conferences. A 3-D electromagnetic model has also been developed, which is coupled with dynamics analysis of the droplets in an electromagnetic levitator, to determine the rotational stability of the droplets. Additional simulations are being conducted and the results are being analyzed. We have also completed the DNS code for direct numerical simulation of turbulence. The code runs in both serial and parallel modes. Some numerical results are obtained and are analyzed. Additional simulations are being carried out. The results obtained so far, though not very conclusive as more calculations are being performed, are consistent with the experimental observations made in space shuttle flights a few years ago. Extensive simulations are being made to further our understanding of the subject.
Bibliography Type:	Description: (Last Updated: 02/07/2014)
Articles in Peer-reviewed Journals	Huo Y, Li BQ. "A mathematical model for Marangoni flow and mass transfer in electrostatically levitated droplets." Metallurgical & Materials Transactions B. 2005 Apr;36(2):271-81. http://dx.doi.org/10.1007/s11663-005-0029-9 , Apr-2005
Articles in Peer-reviewed Journals	Xin A, Li BQ. "Numerical simulation of thermal wave propagation during laser processing of thin films." Journal of Electronic Materials. 2005 May;34(5):583-91. http://dx.doi.org/10.1007/s11664-005-0069-6 , May-2005
Articles in Peer-reviewed Journals	Huo Y, Li BQ. "A finite element-boundary integral method for 3-d electromagnetic heating analysis." ASME Journal of Heat Transfer, in press, as of January 2006. , Jan-2006
Articles in Peer-reviewed Journals	Huo Y, Li BQ. "Surface deformation and convection in electrostatically-positioned droplets of immiscible liquids under microgravity." Journal of Heat Transfer 2006 Jun;128(6):520-9. http://dx.doi.org/10.1115/1.2188460 , Jun-2006
Papers from Meeting Proceedings	Li BQ. "Parallel computing of multiscale transport phenomena during solidification processing." International Symposium on Applied Mathematics, August 2005. International Symposium on Applied Mathematics, August 2005. , Aug-2005
Papers from Meeting Proceedings	Li BQ. "Convection in levitated droplets and its effect on the measurement of thermophysical properties." Interdisciplinary Transport Phenomena in Microgravity and Space Sciences IV, Tomar, Portugal, 7-12 August 2005. Interdisciplinary Transport Phenomena in Microgravity and Space Sciences IV, 2005. p. 3-16, Invited Keynote Speech. , Aug-2005
Papers from Meeting Proceedings	Xu B, Ai X, Li BQ. "Stability of axisymmetric thermocapillary flows in an open vertical cylinder with lateral heating." 2005 ASME Summer Heat Transfer Conference, San Francisco, CA, July 17-22, 2005. 2005 ASME Summer Heat Transfer Conference, 2005. Paper HT2005-72550. , Jul-2005